

1	Course Name:
	EMERGING TRENDS IN AI
2	Course Code:
	IQB21103
3	Year and Semester Offered:
	Year 2 Semester 3
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course explores the latest developments and important trends in Artificial Intelligence (AI). It focuses on building a strong understanding of AI's evolution, current advancements, and future possibilities. Students will learn about key areas like generative AI, AI ethics, and how AI is used in fields such as healthcare, sustainability, finance, cybersecurity, and human-computer interaction. The course also highlights the ethical issues, social impacts, and practical uses of AI technologies.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Apply knowledge of AI's historical development, core concepts, and emerging trends (such as generative AI, edge computing, and NLP) to solve problems in various fields. 2. Discuss the ethical challenges and societal impacts of AI technologies, including issues like bias, fairness, and accountability, to demonstrate adaptability and a commitment to lifelong learning. 3. Demonstrate practical solutions for AI applications across domains (e.g., healthcare, finance, and sustainability) by integrating emerging trends and industry best practices to reflect independent learning and professional responsibility.
8	Content Outline:
	Topic
	1. Introduction to Emerging Trends in AI
	2. Generative AI and Large Language Models (LLMs)
	3. AI Ethics and Responsible AI
	4. AI in Healthcare
	5. AI and Sustainability
	6. AI-Powered Automation
	7. AI in Natural Language Processing (NLP)
	8. AI in Cybersecurity

	9. AI and Human-Computer Interaction (HCI)
	10. AI in Finance
9	References:
	Introduction to AI Safety, Ethics, and Society (1st Edition) Hendrycks, CRC Press (2024) Generative Deep Learning (2nd Edition) David Foster, Orielly (2023)